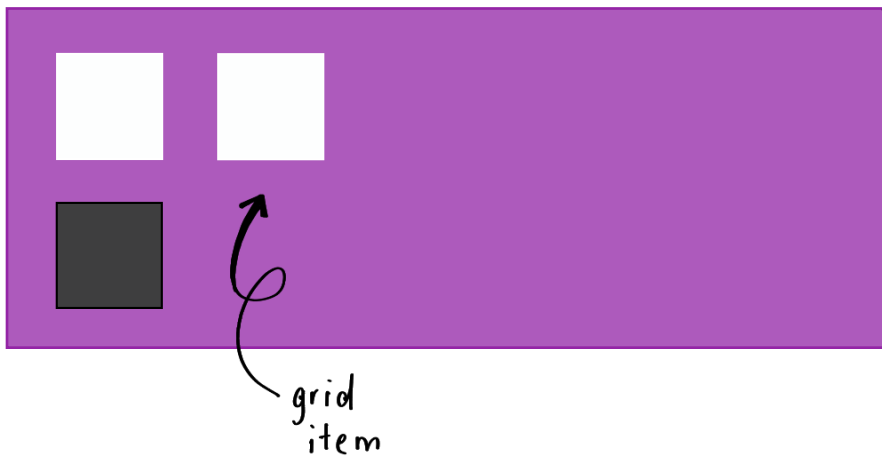


CSS GRID

- Setting a container's display to grid will make all children grid items

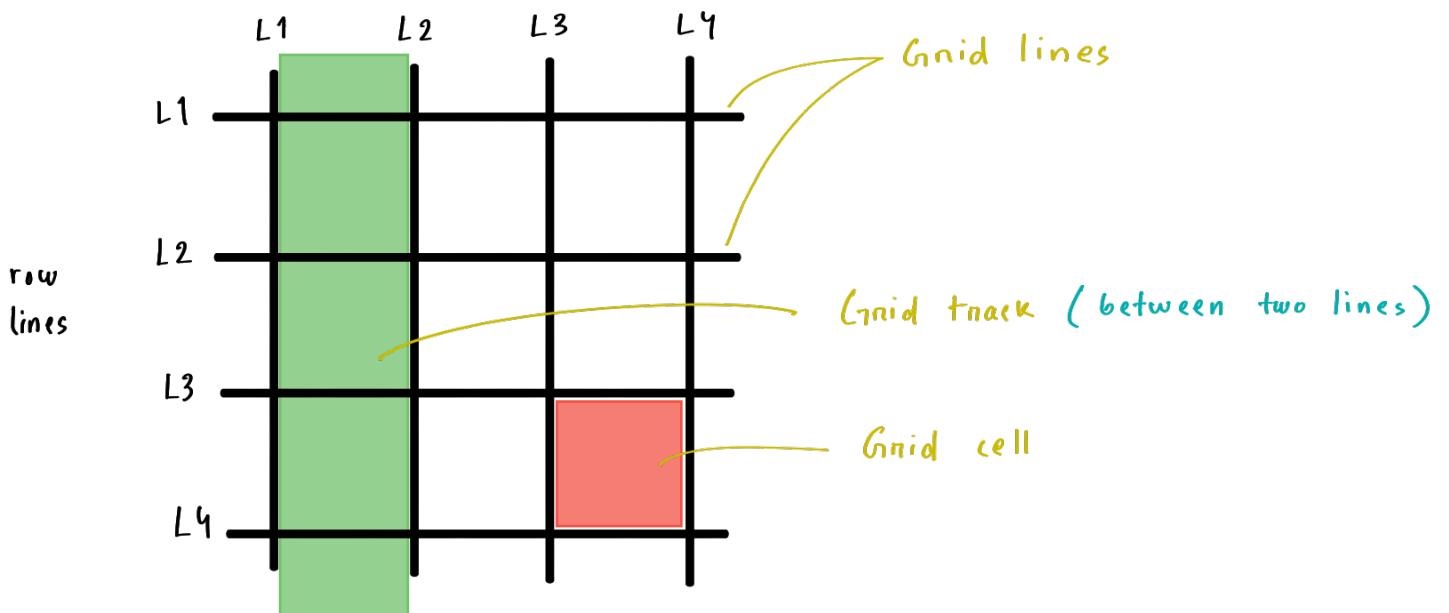
```
Container {  
  display: grid;  
}
```



grid
container

GRID MODEL

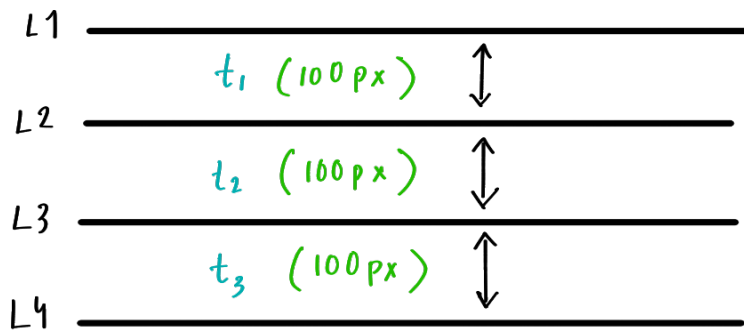
column lines



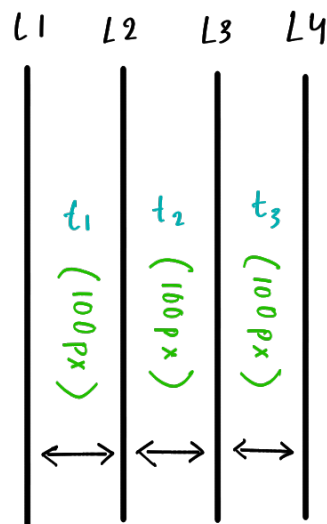
GRID TEMPLATE

— They define the lines and track sizing

- `grid-template-rows : 100px 100px 100px`



- `grid-template-columns : 100px 100px 100px`



- * `grid-template-rows : auto ;` // 1 column
- `grid-template-rows : auto auto ;` // 2 column
- `...`

GRID TEMPLATE

— repeat is used to divide all available space.

- grid-template-rows: repeat (count, 1fr)

no. of rows

1 fraction of 100% available width spaces

OR

grid-template-rows: 1fr 1fr 1fr 1fr ;

- grid-template-columns: repeat (count, 1fr)

GRID GAPS

— They define gaps between the lines

- row-gap: 10px ;

- column-gap: 10px ;

- grid-gap: 10px 20px ; (Shorthand)
 ↓ ↓
 row-gap column-gap

GRID COLUMNS

- Defines an item's starting and ending position inside column

- `grid-column-start : line_number ;`
- `grid-column-end : line_number ;`

Shorthand : `grid-column : start_col / end_col ;`

`grid-column : start_col / span number ;`

↳ no. of cells req.
after starting line

eg: `1 / span 3 ;`

↳ 1

↳ 3 columns
afterwards

GRID ROWS

- Defines an item's starting and ending position inside column

- `grid-row - start : line_number ;`
- `grid-row - end : line_number ;`

Shorthand : `grid-row : start_row / end_row ;`

`grid-row : start_row / span number ;`

COMMON PROPERTIES

- `justify-items (container)`
 - `justify-self (item)`
 - `align-items (container)`
 - `align-self (item)`
 - `place-items`
 - `place-self`
- } horizontal
- } vertical
- } when horizontal,
vertical same
- } Start ;
end ;
center ;



Note

display : inline - grid

(makes grid like an inline element)

Grid → block element

CSS ANIMATIONS

— To animate CSS elements

① keyframes myName {

user defined name

from {

font-size : 20px ;

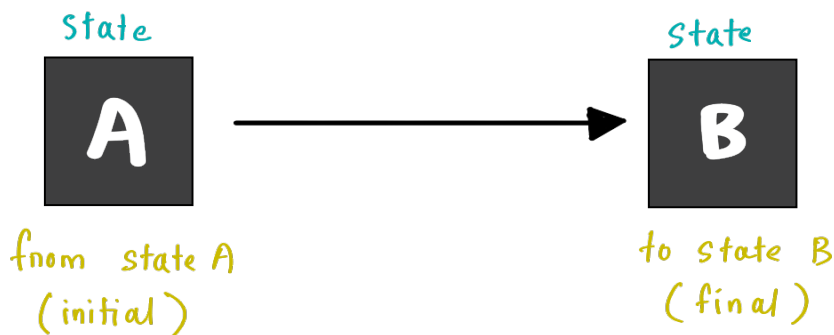
}

to {

font-size : 40px ;

}

}



Properties :

- animation — name
- animation — duration
- animation — timing — function
- animation — delay
- animation — iteration — count
- animation — direction

(no. of times animatⁿ
is performed)

ANIMATION SHORTHAND

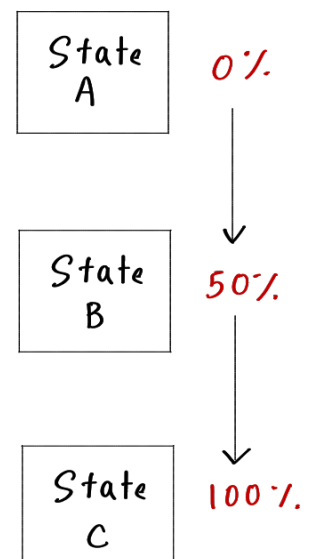
animation :	myName	2s	Linear	3s	infinite	normal ;
	↓ name	↓ duration	↓ timing funct ⁿ	↓ delay	↓ iterat ⁿ count	↓ direct ⁿ

% IN ANIMATION

⑨ keyframes myName {

```
0% { font-size: 20px; }
50% { font-size: 40px; }
100% { font-size: 60px; }
```

}



MEDIA QUERIES

— helps create a responsive website

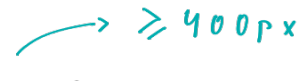
ex : `@media (width : 400px) {`

`{`
 background-color : red ;
 }
}

- `@media (min-width : 400px) {`

`{`
 background-color : red ;
 }
}
 

- `@media (max-width : 400px) {`

`{`
 background-color : red ;
 }
}
 

- `@media (min-width: 400px) and (max-width: 400px) {`

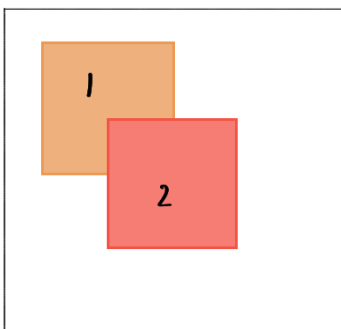
`{`
 `background-color: red;`
 `}`
}

- `@media (orientation: landscape) {`

`{`
 `background-color: red;`
 `}`
}

Z - INDEX

- It decides the stack level of elements
- Overlapping elements with a larger Z-index cover those with a smaller one,



2
1

Z-index : auto (0) → default (same level)

Z-index : 1/2 / ...

Z-index : -1 / -2 / ...

Note : ① higher Z-index → on top

② doesn't work for position - static / default